

Networking Basics

1. What is a Computer Network?

- A) A collection of hardware components and computers
- B) Interconnected by communication channels
- C) Allows sharing of resources and information
- D) All of the Above

Answer: D) All of the Above

2. The first computer network of the world was:

- A) I-net
- B) NSFnet
- C) ARPANET
- D) E-net

Answer: C) ARPANET

3. Which network type is typically used to connect computers in a single office or building?

- A) WAN
- B) LAN
- C) MAN
- D) PAN

Answer: B) LAN

4. What does LAN stand for?

- A) Local Area Network
- B) Large Area Network
- C) Long Area Network
- D) Local Access Network

Answer: A) Local Area Network

5. Which network covers a large geographical area, often a country or continent?

- A) LAN
- B) MAN
- C) WAN
- D) PAN

Answer: C) WAN

6. What does WAN stand for?

- A) Wide Access Network
- B) Wide Area Network
- C) Web Area Network
- D) Wireless Area Network

Answer: B) Wide Area Network

7. A network that connects devices within a city is called:

- A) LAN
- B) MAN
- C) WAN
- D) SAN

Answer: B) MAN (Metropolitan Area Network)

8. Bluetooth is an example of which type of network?

- A) PAN
- B) LAN
- C) VPN
- D) WAN

Answer: A) PAN (Personal Area Network)

9. What does VPN stand for?

- A) Virtual Public Network
- B) Virtual Private Network
- C) Viral Private Network

D) Visual Private Network

Answer: B) Virtual Private Network

10. Which topology requires a central controller or hub?

A) Star

B) Mesh

C) Ring

D) Bus

Answer: A) Star

11. In which topology is every node connected to every other node?

A) Star

B) Mesh

C) Bus

D) Ring

Answer: B) Mesh

12. Which topology uses a single backbone cable to connect all nodes?

A) Star

B) Mesh

C) Bus

D) Tree

Answer: C) Bus

13. In a Ring topology, data travels in:

A) Both directions simultaneously

B) One direction only (unidirectional) or both (bidirectional - dual ring)

C) Random directions

D) No direction

Answer: B) One direction only (unidirectional) or both (bidirectional - dual ring)

14. Which device is used to connect multiple computers in a Star topology?

- A) Repeater
- B) Hub/Switch
- C) Modem
- D) Terminator

Answer: B) Hub/Switch

15. What is the main disadvantage of a Bus topology?

- A) Requires a lot of cable
- B) Difficult to install
- C) If the backbone cable fails, the entire network goes down
- D) Expensive

Answer: C) If the backbone cable fails, the entire network goes down

16. Which network device is known as an intelligent device?

- A) Hub
- B) Switch
- C) Repeater
- D) Cable

Answer: B) Switch

17. A Hub operates in which layer of the OSI model?

- A) Physical Layer
- B) Data Link Layer
- C) Network Layer
- D) Transport Layer

Answer: A) Physical Layer

18. A Switch operates in which layer of the OSI model?

- A) Physical Layer
- B) Data Link Layer

C) Network Layer

D) Application Layer

Answer: B) Data Link Layer

19. A Router operates in which layer of the OSI model?

A) Data Link Layer

B) Network Layer

C) Transport Layer

D) Session Layer

Answer: B) Network Layer

20. Which device connects two networks with different protocols?

A) Bridge

B) Gateway

C) Switch

D) Hub

Answer: B) Gateway

21. What does OSI stand for?

A) Open System Interconnection

B) Open Source Interconnection

C) Optical Source Interconnection

D) Open Service Internet

Answer: A) Open System Interconnection

22. How many layers are there in the OSI model?

A) 4

B) 5

C) 6

D) 7

Answer: D) 7

23. Which layer is responsible for physical addressing (MAC address)?

- A) Physical
- B) Data Link
- C) Network
- D) Transport

Answer: B) Data Link

24. Which layer is responsible for logical addressing (IP address) and routing?

- A) Data Link
- B) Network
- C) Transport
- D) Session

Answer: B) Network

25. Which layer ensures reliable data transfer (end-to-end)?

- A) Physical
- B) Network
- C) Transport
- D) Application

Answer: C) Transport

26. Which layer is responsible for encryption and decryption?

- A) Application
- B) Presentation
- C) Session
- D) Transport

Answer: B) Presentation

27. Which layer provides the interface to the user?

- A) Application

- B) Session
- C) Presentation
- D) Transport

Answer: A) Application

28. Which of the following is NOT a guided medium?

- A) Twisted Pair
- B) Coaxial Cable
- C) Fiber Optic
- D) Radio Waves

Answer: D) Radio Waves

29. Which cable transmits data as light pulses?

- A) Coaxial
- B) Fiber Optic
- C) Twisted Pair
- D) Ethernet

Answer: B) Fiber Optic

30. UTP stands for:

- A) Unshielded Twisted Pair
- B) Uniform Twisted Pair
- C) Unified Transfer Protocol
- D) Unshielded Transfer Path

Answer: A) Unshielded Twisted Pair

31. Which connector is commonly used with UTP cables?

- A) RJ-11
- B) RJ-45
- C) BNC
- D) USB

Answer: B) RJ-45

32. What is the physical address of a network interface card called?

- A) IP Address
- B) MAC Address
- C) DNS Address
- D) Port Address

Answer: B) MAC Address

33. A MAC address is how many bits long?

- A) 32
- B) 48
- C) 64
- D) 128

Answer: B) 48

34. Which device regenerates the signal to extend the network range?

- A) Switch
- B) Bridge
- C) Repeater
- D) Router

Answer: C) Repeater

35. Which device connects two LAN segments of the same type?

- A) Repeater
- B) Bridge
- C) Gateway
- D) Modem

Answer: B) Bridge

36. Modem stands for:

- A) Modulator Demodulator
- B) Modern Demodulator
- C) Modulator Demagnetizer
- D) Mobile Demodulator

Answer: A) Modulator Demodulator

37. Which transmission mode allows communication in both directions but not simultaneously?

- A) Simplex
- B) Half-Duplex
- C) Full-Duplex
- D) Multiplex

Answer: B) Half-Duplex

38. Which transmission mode allows communication in both directions simultaneously?

- A) Simplex
- B) Half-Duplex
- C) Full-Duplex
- D) Complex

Answer: C) Full-Duplex

39. Television broadcasting is an example of which transmission mode?

- A) Simplex
- B) Half-Duplex
- C) Full-Duplex
- D) Automatic

Answer: A) Simplex

40. Walkie-Talkie is an example of:

- A) Simplex
- B) Half-Duplex
- C) Full-Duplex

D) None

Answer: B) Half-Duplex

41. Telephone conversation is an example of:

A) Simplex

B) Half-Duplex

C) Full-Duplex

D) None

Answer: C) Full-Duplex

42. Which is the fastest transmission media?

A) Twisted Pair

B) Coaxial

C) Fiber Optic

D) Wireless

Answer: C) Fiber Optic

43. Which of the following is a private IP address?

A) 12.0.0.1

B) 192.168.1.1

C) 168.192.1.1

D) 8.8.8.8

Answer: B) 192.168.1.1

44. IPv4 addresses are usually represented in:

A) Binary

B) Dotted Decimal

C) Hexadecimal

D) Octal

Answer: B) Dotted Decimal

45. The process of converting digital signals to analog is called:

- A) Modulation
- B) Demodulation
- C) Encoding
- D) Decoding

Answer: A) Modulation

46. What is "Latency" in networking?

- A) The speed of data
- B) The delay in data transmission
- C) The amount of data lost
- D) The width of the cable

Answer: B) The delay in data transmission

47. "Bandwidth" refers to:

- A) The physical width of the cable
- B) The range of frequencies available for transmission
- C) The number of computers on a network
- D) The length of the cable

Answer: B) The range of frequencies available for transmission (Data transfer rate)

48. A firewall is used to:

- A) Cool down the server
- B) Prevent unauthorized access
- C) Connect cables
- D) Boost signal

Answer: B) Prevent unauthorized access

49. Which protocol is used to browse websites?

- A) FTP
- B) HTTP

C) SMTP

D) SNMP

Answer: B) HTTP

50. HTTPS stands for:

A) Hyper Text Transfer Protocol Secure

B) Hyper Text Transfer Protocol Standard

C) Hyper Transfer Text Protocol Secure

D) High Transfer Text Protocol Secure

Answer: A) Hyper Text Transfer Protocol Secure

51. Which protocol is used to upload/download files?

A) HTTP

B) DNS

C) FTP

D) SMTP

Answer: C) FTP (File Transfer Protocol)

52. Which protocol assigns IP addresses automatically?

A) DHCP

B) DNS

C) ARP

D) ICMP

Answer: A) DHCP

53. DNS is used to:

A) Assign IPs

B) Resolve domain names to IP addresses

C) Transfer files

D) Secure the network

Answer: B) Resolve domain names to IP addresses

54. The physical path over which a message travels is:

- A) Protocol
- B) Signal
- C) Medium
- D) Route

Answer: C) Medium

55. In which topology does the failure of one node NOT affect the others?

- A) Bus
- B) Ring (Single)
- C) Star
- D) None of the above

Answer: C) Star

56. What is the unit of data at the Physical Layer?

- A) Bit
- B) Frame
- C) Packet
- D) Segment

Answer: A) Bit

57. What is the unit of data at the Data Link Layer?

- A) Bit
- B) Frame
- C) Packet
- D) Segment

Answer: B) Frame

58. What is the unit of data at the Network Layer?

- A) Bit

- B) Frame
- C) Packet
- D) Segment

Answer: C) Packet

59. What is "Attenuation"?

- A) Strengthening of signal
- B) Loss of signal strength over distance
- C) Speed of signal
- D) Delay

Answer: B) Loss of signal strength over distance

60. Which device creates a separate collision domain for each port?

- A) Hub
- B) Repeater
- C) Switch
- D) Cable

Answer: C) Switch

61. Which device broadcasts data to all ports?

- A) Switch
- B) Router
- C) Hub
- D) Bridge

Answer: C) Hub

62. CSMA/CD is used in:

- A) Token Ring
- B) Ethernet
- C) FDDI
- D) Wireless

Answer: B) Ethernet

63. What does NIC stand for?

- A) Network Internal Card
- B) Network Interface Card
- C) New Internet Connection
- D) Network Interconnect Chip

Answer: B) Network Interface Card

64. Which cable typically uses a BNC connector?

- A) Twisted Pair
- B) Fiber Optic
- C) Coaxial
- D) USB

Answer: C) Coaxial

65. What is "Crosstalk"?

- A) Signal interference from adjacent wires
- B) Talking on the phone
- C) Data collision
- D) Broken cable

Answer: A) Signal interference from adjacent wires

66. Which type of switching is used in the Internet?

- A) Circuit Switching
- B) Packet Switching
- C) Message Switching
- D) Line Switching

Answer: B) Packet Switching

67. Which of the following is a Loopback address?

- A) 127.0.0.1
- B) 192.168.0.1
- C) 10.0.0.1
- D) 255.0.0.0

Answer: A) 127.0.0.1

68. What is a "Node" in a network?

- A) The wire
- B) Any device connected to the network
- C) The internet
- D) A broken link

Answer: B) Any device connected to the network

69. Which command checks connectivity to another computer?

- A) ipconfig
- B) ping
- C) netstat
- D) msconfig

Answer: B) ping

70. "Topology" refers to:

- A) The physical or logical layout of a network
- B) The speed of the network
- C) The type of cable used
- D) The protocol used

Answer: A) The physical or logical layout of a network

71. Which layer manages the dialogue (session) between computers?

- A) Transport
- B) Session
- C) Network

D) Data Link

Answer: B) Session

72. Wireless networks use what for transmission?

A) Light

B) Sound

C) Radio waves / Microwaves

D) Copper

Answer: C) Radio waves / Microwaves

73. What is an Intranet?

A) A global network

B) A private network restricted to an organization

C) A network connecting two cities

D) The Internet

Answer: B) A private network restricted to an organization

74. What is an Extranet?

A) The internet

B) An intranet accessible to authorized outsiders

C) A public network

D) A wireless network

Answer: B) An intranet accessible to authorized outsiders

75. SMTP stands for:

A) Simple Mail Transfer Protocol

B) Secure Mail Transfer Protocol

C) Simple Message Text Protocol

D) System Mail Transfer Protocol

Answer: A) Simple Mail Transfer Protocol

76. Which topology is most expensive to implement?

- A) Bus
- B) Star
- C) Ring
- D) Mesh (Full)

Answer: D) Mesh (Full) (Due to cabling costs)

77. A proxy server is used for:

- A) Speeding up internet access (Caching)
- B) Hiding internal IP addresses
- C) Filtering content
- D) All of the above

Answer: D) All of the above

78. Which device works as a "Traffic Cop" on the network?

- A) Hub
- B) Repeater
- C) Router
- D) Modem

Answer: C) Router

79. The unique identifier burned into the NIC is:

- A) IP Address
- B) Port Number
- C) MAC Address
- D) Socket

Answer: C) MAC Address

80. A standard Ethernet network cable is limited to approximately:

- A) 10 meters
- B) 100 meters

C) 500 meters

D) 1 km

Answer: B) 100 meters

81. Which layer translates data formats (e.g., Encryption, Compression)?

A) Application

B) Presentation

C) Session

D) Transport

Answer: B) Presentation

82. Which protocol is used to send error messages (e.g., Destination Unreachable)?

A) TCP

B) UDP

C) ICMP

D) IP

Answer: C) ICMP

83. WLAN stands for:

A) Wide Local Area Network

B) Wireless Local Area Network

C) World Local Area Network

D) Web Local Area Network

Answer: B) Wireless Local Area Network

84. Which device modulates and demodulates signals?

A) Switch

B) Router

C) Modem

D) Gateway

Answer: C) Modem

85. A collection of networks connected together is called:

- A) Intranet
- B) Internetwork (or Internet)
- C) LAN
- D) Extranet

Answer: B) Internetwork (or Internet)

86. Half-Duplex transmission is similar to:

- A) A two-way street
- B) A one-way street
- C) A one-lane bridge where traffic goes one way at a time
- D) A highway

Answer: C) A one-lane bridge where traffic goes one way at a time

87. Which IP address class allows the most hosts?

- A) Class A
- B) Class B
- C) Class C
- D) Class D

Answer: A) Class A

88. Which standard governs Wireless LANs (Wi-Fi)?

- A) IEEE 802.3
- B) IEEE 802.5
- C) IEEE 802.11
- D) IEEE 802.15

Answer: C) IEEE 802.11

89. Ethernet uses which standard?

- A) IEEE 802.3

- B) IEEE 802.5
- C) IEEE 802.11
- D) IEEE 802.12

Answer: A) IEEE 802.3

90. Data flow between two devices happens in which modes?

- A) Simplex, Half-Duplex, Full-Duplex
- B) Single, Double, Triple
- C) Main, Sub, Omni
- D) Primary, Secondary

Answer: A) Simplex, Half-Duplex, Full-Duplex

91. What is "Noise" in networking?

- A) Loud sounds
- B) Unwanted electrical or electromagnetic energy that degrades signal quality
- C) High bandwidth
- D) Fast data speed

Answer: B) Unwanted electrical or electromagnetic energy that degrades signal quality

92. A "Server" is:

- A) A computer that requests resources
- B) A computer that provides resources and services
- C) The network cable
- D) The internet

Answer: B) A computer that provides resources and services

93. A "Client" is:

- A) A computer that provides resources
- B) A computer that requests resources from a server
- C) The router
- D) The administrator

Answer: B) A computer that requests resources from a server

94. Peer-to-Peer (P2P) network means:

- A) There is one central server
- B) All computers have equal status and can act as both client and server
- C) Computers are connected in a circle
- D) It uses fiber optics

Answer: B) All computers have equal status and can act as both client and server

95. ISP stands for:

- A) Internet Service Provider
- B) Internal Service Protocol
- C) Internet System Program
- D) International Service Point

Answer: A) Internet Service Provider

96. Which cabling is immune to Electromagnetic Interference (EMI)?

- A) Coaxial
- B) UTP
- C) Fiber Optic
- D) STP

Answer: C) Fiber Optic (It uses light)

97. What is the purpose of a Subnet Mask?

- A) To hide the IP address
- B) To identify the network and host portions of an IP address
- C) To encrypt data
- D) To speed up the internet

Answer: B) To identify the network and host portions of an IP address

98. Which layer is responsible for breaking data into segments?

A) Application

B) Transport

C) Network

D) Physical

Answer: B) Transport

99. In a Star topology, if the central hub fails:

A) Only one computer fails

B) Half the network fails

C) The entire network fails

D) The network works slower

Answer: C) The entire network fails

100. URL stands for:

A) Uniform Resource Locator

B) Universal Resource Link

C) Uniform Reference Line

D) United Resource Locator

Answer: A) Uniform Resource Locator